HW 3 Solution: Three beams under distributed loads

Given data:

$$\mathrm{L} = 450\ \mathrm{mm},\ \mathrm{b} = 16\ \mathrm{mm},\ \mathrm{h} = 8\ \mathrm{mm},\ \mathrm{E} = 2200\ \mathrm{N/mm^2}$$

Therefore:

$$A = bh = 128\ mm^2\ ,\mathrm{I} = \frac{\mathrm{bh}^3}{12} = 682.67\ \mathrm{mm}^4;\ \ \mathrm{EI} = 1{,}501{,}866.67\ \mathrm{N\ mm^2}$$

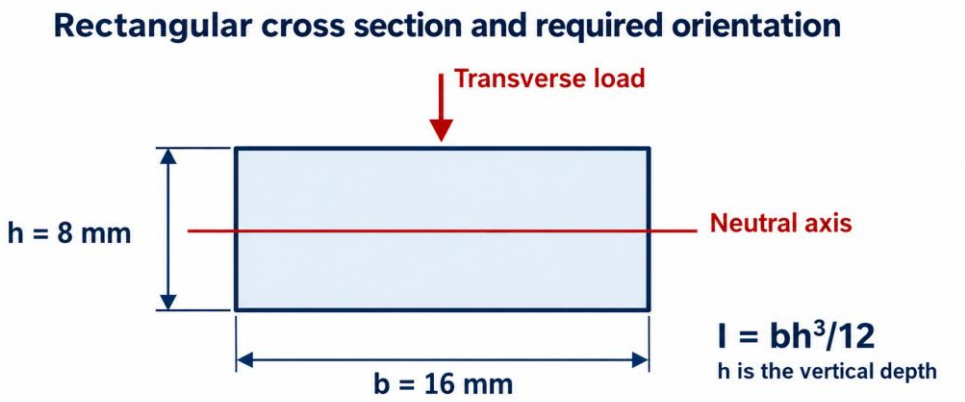


Figure 1. Rectangular cross section used for all three beams. The 8 mm dimension is vertical and parallel to the applied load.

R is positive downward. C is an external support couple, positive clockwise. The distributed load w and displacement v are positive downward; θ = dv/dx is positive clockwise. Internal M is positive in sagging. A is the left end, and B is the right end.

A1 Resultant force and location

$$F_h = \int_0^L w(x)\,dx,\ \ M_{load,A} = \int_0^L x\,w(x)\,dx,\ \ \bar{x} = \frac{M_{load,A}}{F_h}$$

$M_{load,A}$ is the clockwise moment of the downward distributed load about A. It is an applied-load moment, not the internal bending moment M(x). All distances are measured from the left end A; the right end B is at x = L.

Case 1: Triangle increasing toward B

$w_t = 0.002\ N/mm$

$$w(x) = w_t\,\frac{x}{L}$$

$$F_h = \int_0^L w_t\,\frac{x}{L}\,dx = \frac{w_t\,L}{2} = 0.45\ N$$

$$M_{load} = \int_0^L x\,w_t\,\frac{x}{L}\,dx = \frac{w_t\,L^2}{3} = 135\ N\cdot mm$$

$$\bar{x} = \frac{M_{load}}{F_h} = \frac{2L}{3} = 300\ mm$$

Case 2: Trapezoidal load

$w_r = 0.001\ N/mm, w_t = 0.002\ N/mm$

$$w(x) = w_r + (w_t - w_r)\frac{x}{L}$$

$$F_h = \int_0^L w(x)\,dx = \frac{L(w_r + w_t)}{2} = 0.675\ N$$

$$M_{load} = \int_0^L x\,w(x)\,dx = \frac{L^2(w_r + 2w_t)}{6} = 168.75\ N \cdot mm$$

$$\bar{x} = \frac{L(w_r + 2w_t)}{3(w_r + w_t)} = 168.75/0.675 = 250\ mm$$

Case 3: Triangle decreasing toward B

$w_0 = 0.002\ N/mm$

$$w(x) = w_0\,(1 - x/L)$$

$$F_h = \int_0^L w_0\,(1 - x/L)\,dx = w_0 L - \frac{w_0 L}{2} = \frac{w_0\,L}{2} = 0.45\ N$$

$$M_{load} = \int_0^L x\,w_0\,(1 - x/L)\,dx = \frac{w_0 L^2}{2} - \frac{w_0 L^2}{3} = \frac{w_0\,L^2}{6} = 67.5\ N \cdot mm$$

$$\bar{x} = \frac{M_{load}}{F_h} = \frac{L}{3} = 150\ mm$$

A2 Equilibrium and support reactions

Case 1: Cantilever

$$\Sigma F_x = 0 \Rightarrow R_{Ax} = 0$$

$$\Sigma F_y = 0: \quad R_A + 0.450 = 0 \Rightarrow R_A = -0.45\ N$$

$$\Sigma M_a = 0: \quad C_A + 0.45(300) = 0$$

$$C_A = -135\ N \cdot mm$$

A supplies an upward force and a counterclockwise couple. B is free, so it has no support reaction. The horizontal reaction at A is zero.

Case 2: Left roller and right pin

$$C_A = C_B = 0,\ \Sigma F_x = 0 \Rightarrow R_{Ax} = R_{Bx} = 0$$

$$\Sigma M_a = 0: \quad 450R_B + 0.675(250) = 0$$

$$R_B = -0.375\ N$$

$$\Sigma F_y = 0: \quad R_A + R_B + 0.675 = 0 \Rightarrow R_A = -0.675 - R_B = -0.3\ N$$

Both reactions act upward. The roller provides no horizontal force; horizontal equilibrium gives zero horizontal reaction at the pin.

Case 3: Fixed at both ends

$$\Sigma F_y = 0: \quad R_A + R_B = -0.45$$

$$\Sigma M_a = 0: \quad C_A + C_B + 450R_B + 67.5 = 0$$

Two equilibrium equations cannot determine four transverse reactions. The two compatibility equations derived in A3 give:

$$R_A = -0.315\,N, \quad R_B = -0.135\,N$$

$$C_A = -20.250\,N\cdot mm, \quad C_B = +13.500\,N\cdot mm$$

Horizontal reactions are zero for this linear bending problem with no axial loading or imposed axial deformation.

$$\Sigma F_{\mathrm{x}} = 0 \Rightarrow R_{Ax} = R_{Bx} = 0$$

A3 Moment equations and integration

$$EI\,v'' = -M(x), \;\; \theta = \frac{dv}{dx}, \qquad V = dM/dx, \;\; dV/dx = -w(x)$$

Cut the beam at x and consider the left segment. Each load element $w(\xi)d\xi$ has a lever arm x − ξ to the cut.

Case 1: Equilibrium is sufficient

No additional compatibility equation is needed to determine reactions.

$$w(x) = w_t\,\frac{x}{L}$$

$$M(x) = C_A - R_A x - \int_0^x w(\xi)(x-\xi)\,d\xi$$

$$M(x) = -\frac{w_t L^2}{3} + \frac{w_t L x}{2} - \frac{w_t x^3}{6L}$$

$$EI\,\theta(x) = +\frac{w_t L^2 x}{3} - \frac{w_t L x^2}{4} + \frac{w_t x^4}{24L} + D_1$$

$$EI\,v(x) = +\frac{w_t L^2 x^2}{6} - \frac{w_t L x^3}{12} + \frac{w_t x^5}{120L} + D_1 x + D_2$$

At the clamp, $v(0) = \theta(0) = 0$:

$$D_1 = 0, \qquad D_2 = 0$$

Case 2: Integration for the end rotations

$$\mathrm{w(x)} = \mathrm{w_r} + kx \;\;, k = \frac{(\mathrm{w_t} - \mathrm{w_r})}{L}$$

$$M(x) = -R_A x - \int_0^x w(\xi)(x-\xi)\,d\xi$$

$$M(x) = -R_A x - \frac{w_r x^2}{2} - \frac{kx^3}{6}$$

$$EI\,v'' = +R_A x + \frac{w_r x^2}{2} + \frac{kx^3}{6}$$

Integrate term by term, using $\int x^{\mathrm{n}}\,dx = x^{\mathrm{n}+1}/(n+1)$. Since EI is constant, it stays on the left.

$$EI\,\theta(x) = +\frac{R_A x^2}{2} + \frac{w_r x^3}{6} + \frac{kx^4}{24} + D_1$$

$$EI\,v(x) = +\frac{R_A x^3}{6} + \frac{w_r x^4}{24} + \frac{kx^5}{120} + D_1 x + D_2$$

The roller and pin impose $v(0) = v(L) = 0$. They allow rotation, so neither end slope is prescribed as zero.

$$v(0) = 0 \;\Rightarrow\; D_2 = 0$$

$$v(L) = 0 \;\Rightarrow\; D_1 = -\frac{R_A L^2}{6} - \frac{w_r L^3}{24} - \frac{kL^4}{120}$$

$$D_1 = 10125 - 3796.875 - 759.375 = 5568.75\ N\,mm^2$$

So, at $x_A = 0$:

$$EI\ \theta(x) = +\frac{R_A x^2}{2} + \frac{w_r x^3}{6} + \frac{kx^4}{24} + D_1$$

$$EI\ \theta(x_A) = D_1$$

$$\theta_A = \frac{5568.75}{1{,}501{,}866.67} = +0.0037\ rad$$

at $x_B = L = 450$:

$$\theta_B = \frac{-30375 + 15187.5 + 3796.875 + 5568.75}{1{,}501{,}866.67}$$

$$\theta_B = -0.0039\ rad$$

Case 3: Compatibility at both clamps

$$w(x) = w_0\,(1 - x/L)$$

$$M(x) = C_A - R_A x - \int_0^x w(\xi)(x - \xi)\,d\xi$$

$$M(x) = C_A - R_A\,x - \frac{w_0 x^2}{2} + \frac{w_0 x^3}{6L}$$

$$EI\ v'' = -C_A + R_A x + \frac{w_0\,x^2}{2} - \frac{w_0\,x^3}{6L}$$

Integrate term by term:

$$\mathrm{EI}\ \theta(x) = -\mathrm{C_A x} + \frac{\mathrm{R_A x^2}}{2} + \frac{\mathrm{w_0\,x^3}}{6} - \frac{\mathrm{w_0\,x^4}}{24\mathrm{L}} + \mathrm{D_1}$$

$$\mathrm{EI\ v(x)} = -\frac{\mathrm{C_A x^2}}{2} + \frac{\mathrm{R_A x^3}}{6} + \frac{\mathrm{w_0\,x^4}}{24} - \frac{\mathrm{w_0\,x^5}}{120\mathrm{L}} + \mathrm{D_1 x} + \mathrm{D_2}$$

At A, $\theta(0) = 0$ gives $D_1 = 0$ and $v(0) = 0$ gives $D_2 = 0$.

$$\theta(0) = \mathrm{v}(0) = 0 \;\Rightarrow\; \mathrm{D_1 = D_2 = 0}$$

At B, the beam must return to its original height and have zero slope. Substituting x = L therefore gives:

$$\theta(\mathrm{L}) = 0 \;\Rightarrow\; -\mathrm{C_A L} + \frac{\mathrm{R_A L^2}}{2} + \frac{\mathrm{w_0\,L^3}}{8} = 0$$

$$\mathrm{v(L)} = 0 \;\Rightarrow\; -\frac{\mathrm{C_A L^2}}{2} + \frac{\mathrm{R_A L^3}}{6} + \frac{\mathrm{w_0\,L^4}}{30} = 0$$

$$-450\mathrm{C_A} + 101250\mathrm{R_A} + 22781.250 = 0$$

$$-101250\mathrm{C_A} + 15187500\mathrm{R_A} + 2733750 = 0$$

Multiply the first equation by L/2 and subtract the second. The $\mathrm{C_A}$ terms cancel:

$$\frac{R_A L^3}{12} + \frac{7w_0 L^4}{240} = 0$$

$$R_A = -\frac{7w_0 L}{20} = -0.315\ \text{N}, \quad C_A = -\frac{w_0 L^2}{20} = -20.250\ \text{N} \cdot \text{mm}$$

Return to the two whole-beam equilibrium equations to obtain the right reactions:

$$R_B = -0.450 - (-0.315) = -0.135\ \text{N}$$
$$C_B = -67.50 - 450(-0.135) - (-20.25) = +13.50\ \text{N mm}$$

Both forces are upward. The left external couple is counterclockwise, and the right external couple is clockwise. The internal end moments are $M(0^+) = -20.25$ N mm and $M(L^-) = -13.50$ N mm.

A4 Displacement and rotation at supports and selected comparison points

Using the integrated elastic-curve equations with EI = 1.50187×10^6 N·mm² gives the following selected values:

Case 1:

$$EI\ \theta(x) = +\frac{w_t L^2 x}{3} - \frac{w_t L x^2}{4} + \frac{w_t x^4}{24L}$$
$$EI\ v(x) = +\frac{w_t L^2 x^2}{6} - \frac{w_t L x^3}{12} + \frac{w_t x^5}{120L}$$

$\theta(L) = w_t L^3/(8EI) = 0.015\ rad$ and $v(L) = 11 w_t L^4/(120EI) = 5.006\ mm$.

Case 2:

$$EI\ \theta(x) = +\frac{R_A x^2}{2} + \frac{w_r x^3}{6} + \frac{k x^4}{24} + D_1$$
$$EI\ v(x) = +\frac{R_A x^3}{6} + \frac{w_r x^4}{24} + \frac{k x^5}{120} + D_1 x$$

$v(L/2) = 0.533\ mm$,

$\theta(0) = 0.0037\ rad$

$\theta(L) = -0.0039\ rad$.

Case 3:

$$EI\ v(x) = -\frac{C_A x^2}{2} + \frac{R_A x^3}{6} + \frac{w_0 x^4}{24} - \frac{w_0 x^5}{120L}$$

$v(L/2) = w_0 L^4/(768EI) = 0.071\ mm$.

Case	Support	v in mm	θ in rad	Interpretation
1	A fixed	0	0	Translation and rotation restrained
2	A roller	0	+0.0037	Clockwise rotation
2	B pin	0	−0.0039	Counterclockwise rotation
3	A fixed	0	0	Translation and rotation restrained
3	B fixed	0	0	Translation and rotation restrained

At a fixed support, v = 0 and θ = 0. At a pin or roller, v = 0 while θ is generally nonzero. The additional free-end and midspan values below make the effect of the specified section stiffness EI visible in every case.

B Software solution and comparison

Each support and selected deformation location has analytical and software results. R is positive downward, C is positive clockwise, v is positive downward, and θ is positive clockwise. Software results are rounded to the displayed precision.

Case and support	Method	R in N	C in N mm	v in mm	θ in rad
1 A	Hand	−0.45	−135	0	0
1 A	Software	−0.45	−135	0	0
2 A	Hand	−0.3	0	0	+0.0037
2 A	Software	−0.3	0	0	+0.00370789
2 B	Hand	−0.375	0	0	−0.0039
2 B	Software	−0.375	0	0	−0.00387643
3 A	Hand	−0.315	−20.25	0	0
3 A	Software	−0.315	−20.25	0	0
3 B	Hand	−0.135	+13.5	0	0
3 B	Software	−0.135	+13.5	0	0

Case 1: Cantilever

STATION	FORCE (+DOWN)	MOMENT (+CLOCKWISE)	V (+DOWN)	Θ = DV/DX
0 mm	-0.450000 N	-135.000000 N·mm	0.000000 mm	0.00000000 rad

Case 2: Left roller and right pin

STATION	FORCE (+DOWN)	MOMENT (+CLOCKWISE)	V (+DOWN)	Θ = DV/DX
0 mm	-0.300000 N	0.000000 N·mm	0.000000 mm	0.00370789 rad
450 mm	-0.375000 N	0.000000 N·mm	0.000000 mm	-0.00387643 rad

Case 3: Fixed at both ends

STATION	FORCE (+DOWN)	MOMENT (+CLOCKWISE)	V (+DOWN)	Θ = DV/DX
0 mm	-0.315000 N	-20.250000 N·mm	0.000000 mm	0.00000000 rad
450 mm	-0.135000 N	13.500000 N·mm	0.000000 mm	0.00000000 rad

All horizontal reactions are zero analytically. The software uses a transverse-bending model and has no independent axial reaction output.

Difference and numerical checks

The analytical and numerical results agree after rounding to the displayed analytical precision.

Selected deformation checks

Positive v is downward, and positive θ is clockwise. Software values should agree with the analytical results to the displayed precision.

Case	Location	Quantity	Analytical	Software
1	Free end, x = L	v	5.006 mm	5.00565 mm
1	Free end, x = L	θ	0.015 rad	0.0151686 rad
2	Midspan, x = L/2	v	0.533 mm	0.533272 mm
2	Left support, x = 0	θ	0.0037 rad	0.00370789 rad
2	Right support, x = L	θ	−0.0039 rad	−0.00387643 rad
3	Midspan, x = L/2	v	0.071 mm	0.0711029 mm